



BLM4140 – Wireless & Mobile Networks
Wi-Fi Performance Analysis Project

Teacher: Dr. Mehmet Sukru KURAN

Student: Melih ALCIK(22011628)

Video Link: <https://youtu.be/shF8tUHDuqI>

What I Did (Simulation Setup)

For this project, I used the NS-3 simulator on Ubuntu to test how an IEEE 802.11n network handles different traffic loads. Following the correction that 802.11ac doesn't actually support the 2.4 GHz frequency, I configured the PHY layer to standard 802.11n for the 2.4 GHz band.

I set up a single Access Point (AP) and tested it with 4, 8, 12, and 20 stations (STAs). I positioned all the STAs right next to the AP so that bad signal strength wouldn't mess with the data; I just wanted to test pure network congestion as listed in the project. To hit the exact target loads (50%, 80%, and 90% of our max 72.2 Mbps raw capacity for 11n), I used an OnOffApplication over pure uplink TCP traffic. I also made sure modern MAC features like EDCA, TXOP, Block ACK, and Frame Aggregation were turned on for all the tests to see how they compared against using the RTS/CTS handshake.

RESULTS

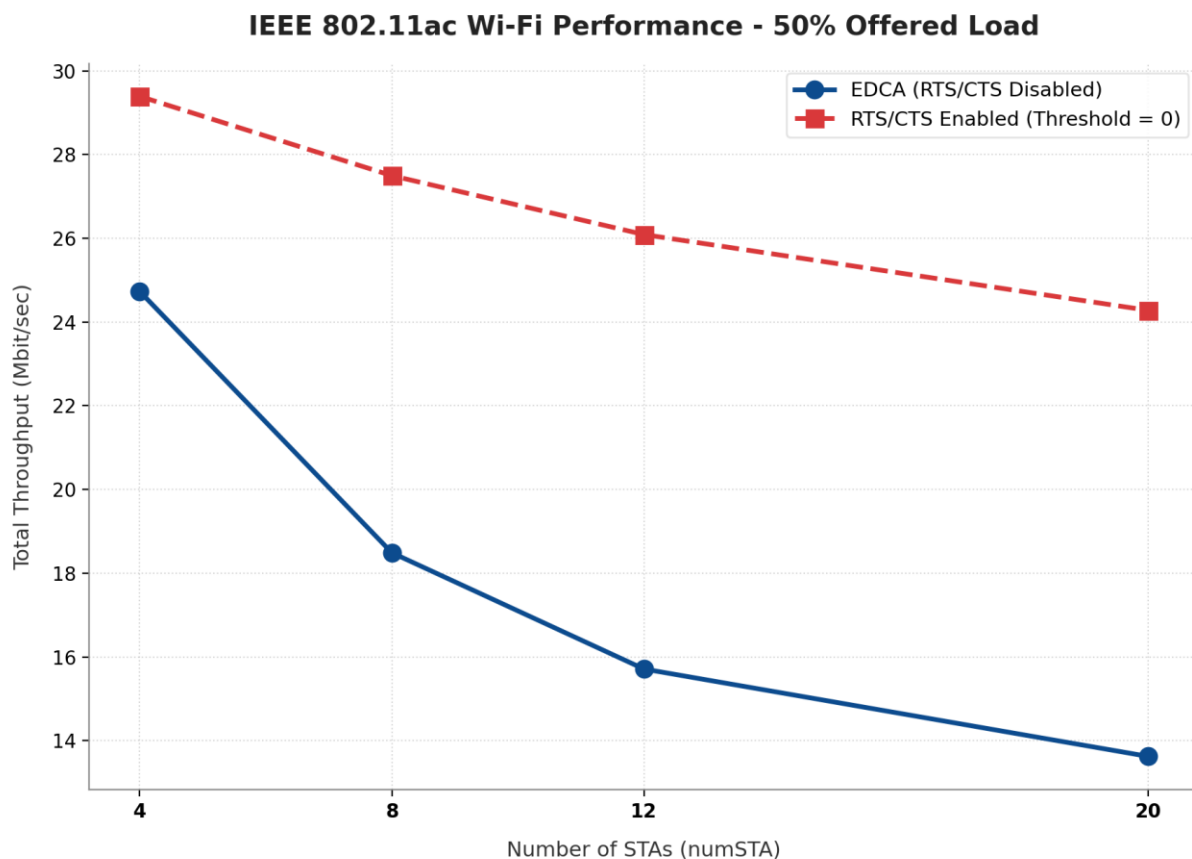


Figure 1 (Results for 50% Load)

Analysis: At 50% load, the network isn't under too much stress yet. The standard EDCA setup (the blue line) handles the traffic fine, dropping smoothly from 24.7 Mbps at 4 STAs down to 13.6 Mbps at 20 STAs. However, the RTS/CTS mechanism (the red line) is noticeably faster across the board. Usually, asking for permission (RTS) before sending every single packet wastes time and slows small networks down. But because we have Frame Aggregation turned on, the STAs reserve the channel once and then blast a massive chunk of data at the AP. That makes the RTS overhead totally worth it, even at a medium load.

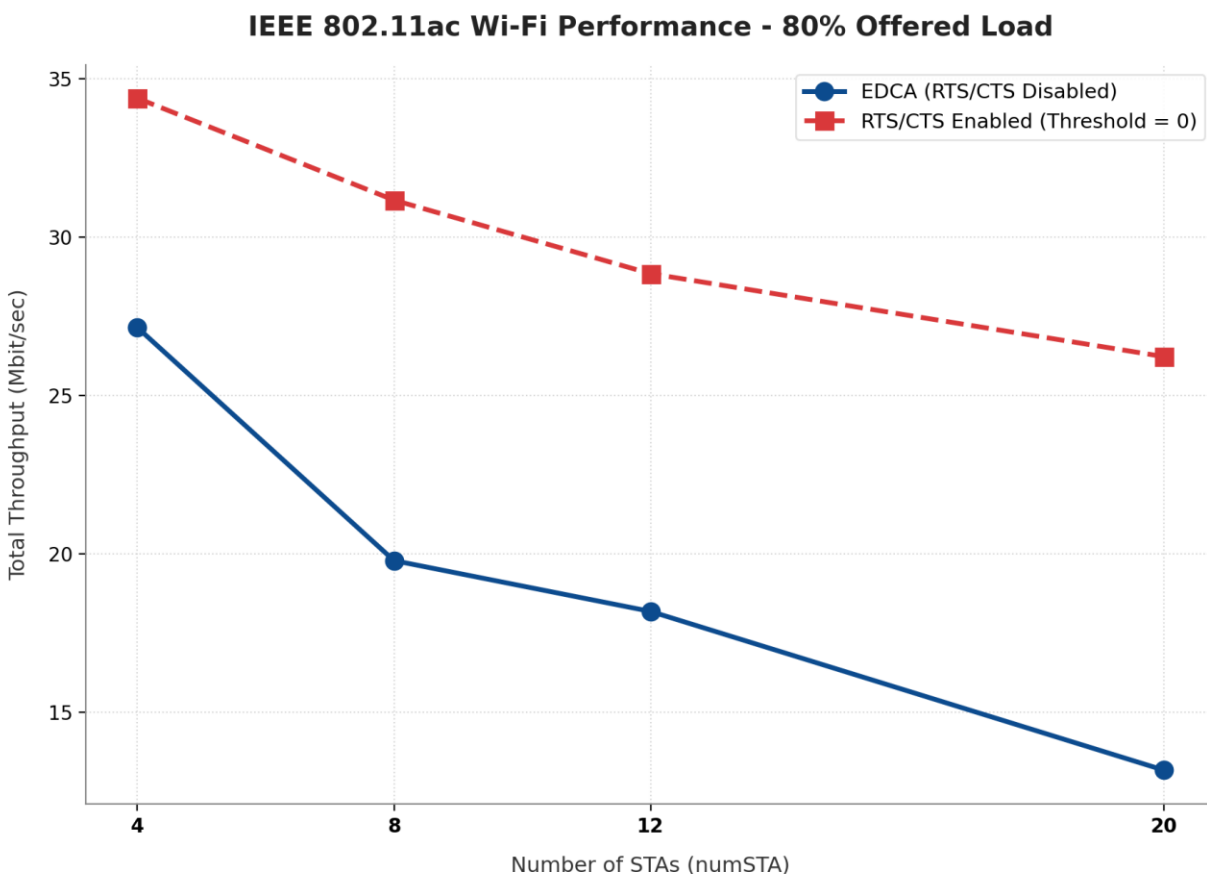


Figure 2(Results for 80% Load)

Analysis: When we push the network to 80% capacity, the difference becomes a lot more obvious. The standard EDCA curve starts dropping really fast after we pass 8 stations. By the time we hit 20 STAs, it's down to 13.1 Mbps. This happens because the air is getting crowded, and the devices are constantly talking over each other, colliding, and having to wait to try again. RTS/CTS, on the other hand, handles the crowd much better. Because the AP explicitly tells nodes when it's safe to talk, we avoid all those

collisions and keep a solid throughput of 26.2 Mbps at 20 STAs.

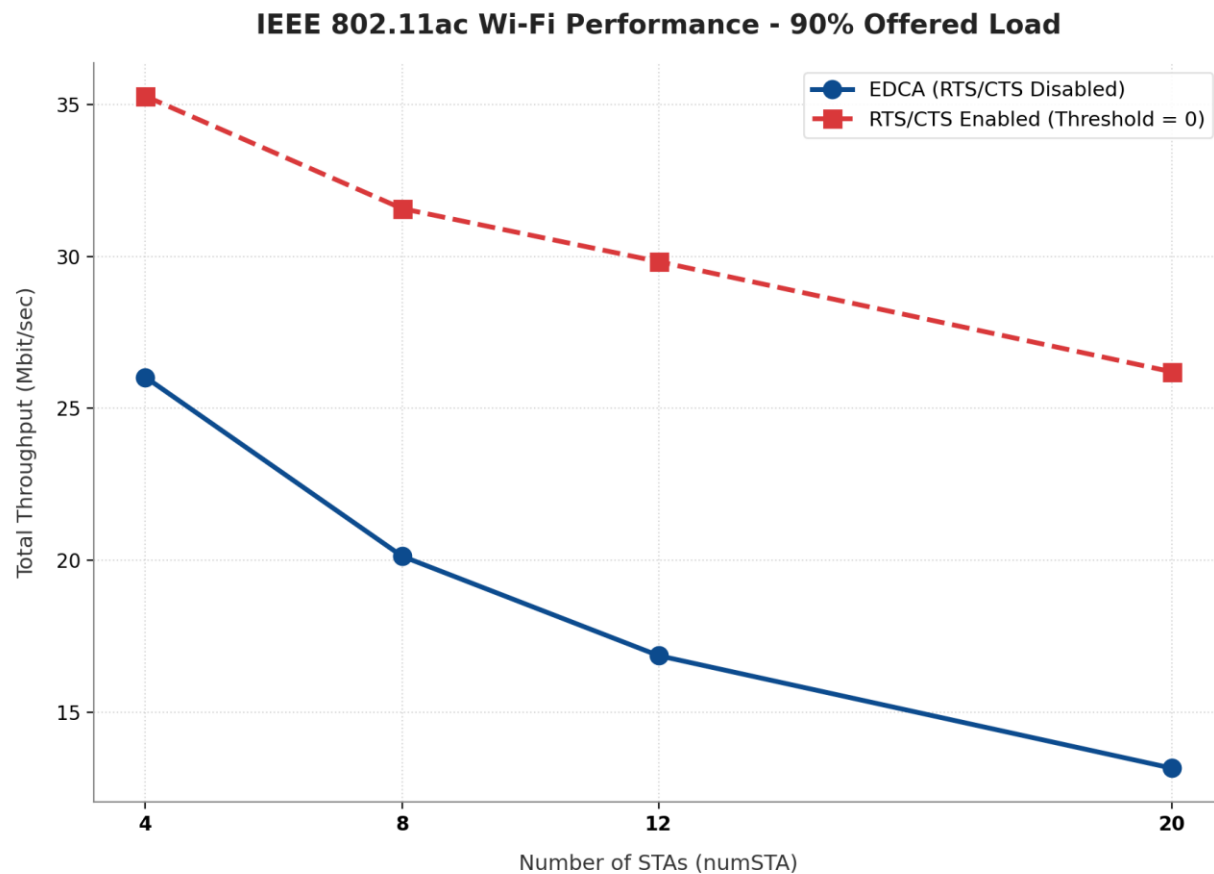


Figure 3(Results for 90% Load)

Analysis: This was the real stress test, and it actually showed the "CSMA scaling" problem mentioned in your recent Wi-Fi QoS Management paper [1]. At 90% load with 20 stations, the standard EDCA network basically crashes, hitting a wall at 13.1 Mbps. The stations are caught in a loop: they collide, double their wait time, and collide again. They spend more time waiting than actually sending data. But the RTS/CTS mechanism completely fixes this. By using the CTS broadcast to silence hidden nodes, the network doesn't waste time recovering from crashes. It holds strong at 26.1 Mbps, proving that in a really crowded Wi-Fi network, you absolutely need a strict channel reservation handshake.

References

[1] F. Canbal, Y. B. Özgün, M. Ş. Kuran, G. Venkatesan, and N. Canpolat, "Wi-Fi QoS Management Program: Bridging the QoS Gap of Multimedia Traffic in Wi-Fi Networks," *IEEE Communications Magazine*, 2024.

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